

Course code: ME411

Course Name: Finite Elements Method

Pre-requisite: Strength of Materials

Objective of the course:

Solving real field problem has been made easy with the advent of Finite Element Method (FEM). FEM as a numerical tool is applicable to almost all field of Engineering and Science, hitherto either experimental set ups are much more prone to errors or such unrealistic assumptions of Analytical methods.

Course Content:

UNIT I: COMPUTATIONAL MODELING(5 Lectures)

Introduction to numerical methods, Physical Problems in Engineering, Three-dimensional stress analysis: 3 D Solid, Plates, Beams, Truss, Computational modeling using FEM, Commercial FEM software and applications, Simulation and Visualization.

UNIT II: MATRIX TECHNIQUES(5 Lectures)

Introduction concept of matrices, Solution of large systems of algebraic equations, discretization of domain: Type of elements. Shape functions. Isoparametric elements.

UNIT III: VARIATIONAL METHODS (6 Lectures)

Variational formulation. Euler - Lagrange equation. Introduction to δ operator and its use for solution of heat conduction problems. Methods of weighted residual techniques-Weak Formulation, Galerkin FE formulation, Rayleigh & Ritz FEM.

UNIT IV: FEM FOR TRUSSES, BEAMS AND FRAMES (7 Lectures)

FEM analysis of One-dimensional problems:-Second order boundary value problem, Applications in Solid, Fluid and Heat Transfer, Bending of Beams, Euler-Bernoulli Beam, Truss/Frame analysis, 2D Problems

UNIT V: EIGENVALUE AND TIME DEPENDENT PROBLEMS (7 Lectures)

Eigen problems, Transient Dynamic Problems, Hamilton's Principle, Free vibration analysis and methods of solution.

UNIT VI: NUMERICAL INTEGRATION AND APPLICATION (8 Lectures)

Numerical Integration and Computer Implementation-Natural Co-ordinates, Interpolation Functions, Numerical Integration, Isoparametric Formulation, Computer Implementation, Applications in two-dimensional problems-Plane Stress and plane Strain, Three Dimensional Solid, Axi-symmetric Solid, Plate and Shell structures, Flow of Viscous Fluid, MATLAB Application,

UNIT VII: Special Topics (5 Lectures)

Stationary Singular elements, Quarter-Point Singular Elements, Moving singular elements, Semi-infinite elements, Buckling analysis, Non-linear analysis

Laboratory Component: No

Readings:

Reference Books:

1. Finite Element Analysis: B. Izabo and I Babuska
2. The Finite Element Method for Solid and structural Mechanics: O.C. Zienkiewicz and R. L. Taylor
3. Finite Element Procedures in Engineering Analysis: K. J. Bathe
4. An Introduction to Finite Element Method: J. N. Reddy